

AIFP-1 — AiFinPay Payment Protocol Specification

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This is **Document 1 of 4** in the official AiFinPay documentation set:

1. **AIFP-1 — Payment Protocol Specification** *(this document — the normative standard)*
2. [Merchant Integration Guide](#) — server-side integration for 15 frameworks
3. [AI Agent SDK Specification](#) — client-side agent SDK across 7 languages
4. [Security & Cryptography Specification](#) — threat model & crypto

This document is **self-contained**. It defines the protocol normatively. The other three documents are *implementation guidance* that conform to this specification. Where this document and another disagree, **this document governs**.

Status of This Memo

This document specifies the AiFinPay Payment Protocol, version 1 (**AIFP-1**), an application-layer payment protocol layered on top of HTTP. It is published for the AiFinPay developer community, protocol implementers, enterprises, standards bodies, and investors. Distribution is unlimited.

AIFP-1 is a **Draft Standard**. It is stable enough for production implementation and is in active use, but normative details MAY be refined through the Open Governance process (Section 22) prior to promotion to **Internet Standard**. Implementers SHOULD track the protocol changelog (Appendix D) and the **AIFP-Version** negotiation mechanism (Section 8.3).

This memo does not represent the position of any standards-development organization. It is an open specification; implementations MAY compete.

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Abstract

The web is entering an **AI-first** phase in which the primary consumer of content and APIs is no longer a human browser but an autonomous software agent. The two monetization models that built the human web — advertising and subscriptions — both fail for machine consumers: agents do not view ads, cannot complete human registration or KYC, and cannot subscribe to the thousands of resources a single task touches. Traditional payment rails (cards, ACH, SEPA) carry a minimum economically viable transaction of roughly **\$0.30–\$0.50** due to fixed fees, while a typical API request is worth **USD 0.00001–USD 0.00010** — a gap of three to four orders of magnitude.

AIFP-1 inverts the prevailing "block the bots" posture into "monetize the machines." It defines a complete, HTTP-native payment handshake built on the long-reserved **402 Payment Required** status code: a merchant returns a machine-readable **Payment Challenge**, the agent pays through AiFinPay, receives a cryptographically signed **Receipt Token**, replays its original request with the receipt attached, and the merchant grants access after **stateless local signature verification** — no backend round-trip, no human in the loop.

This specification defines the protocol's message formats, HTTP semantics, authentication and authorization, the merchant/agent/wallet flows, x402 compatibility and migration, onboarding, error taxonomy, retry strategy, security and threat model, replay protection, cryptography, receipt verification, idempotency, settlement, multi-chain support (12 networks), fiat support, and the normative requirements an implementation **MUST** satisfy to claim conformance.

Requirements Notation

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in [RFC 2119](#) and [RFC 8174] when, and only when, they appear in all capitals, as shown here.

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1. Vision

1.1. Why the protocol exists

For the first time in the history of the web, the primary consumer of an HTTP resource is increasingly **not a human**. LLM agents, retrieval-augmented-generation (RAG) pipelines, training-data crawlers, and autonomous task executors now generate a large and growing share of API and content traffic.

The human web monetizes through two models, both of which break under machine consumption:

1. **Advertising.** An agent does not watch ads, does not click banners, and does not convert into a buyer. The model produces **zero revenue** from machine traffic.

2. **Subscriptions.** An agent cannot register, has no inbox to confirm, does not pass individual KYC, and cannot economically subscribe to each of the thousands of resources a single task run touches.

Lacking a payment mechanism, providers default to **defense**: blocking via `robots.txt`, WAF fingerprinting, IP rate-limiting, and CAPTCHAs. This is a **negative-sum** outcome — the provider loses revenue, the agent loses access, and infrastructure burns resources fighting traffic it could have monetized.

1.2. The market gap

Problem	Current "solution"	Consequence
Crawlers download training data	<code>robots.txt</code> , legal letters	Data still taken, no revenue
No way to charge a machine a micropayment	Subscriptions, manual API keys	Does not scale to millions of agents
Payment rails are built for humans	Card checkout	Minimum fee exceeds the value of a request
Agents have no identity	—	Payment cannot be attributed

1.3. The AIFP solution

AIFP **monetizes** machine traffic instead of blocking it. Baseline scenario:

1. The merchant installs the AiFinPay middleware (a single line).
2. The merchant configures pricing: the first **100 requests free** (configurable), then **USD 0.00001–USD 0.00010** per request by pricing_tier.
3. On quota exhaustion, the server returns **402 Payment Required** with a machine-readable Payment Challenge.
4. The agent **automatically** pays through AiFinPay.
5. The merchant receives funds (on-chain or fiat/stablecoin settlement).
6. The agent receives a signed **Receipt Token**.
7. The agent **replays** its original request with the receipt attached.
8. The merchant **verifies the receipt locally** by signature — no round-trip.
9. **Access is granted.**

No registration, no subscriptions, no human in the loop — completed in a single `challenge → pay → retry` cycle.

1.4. The machine economy

AIFP lays rails for a **Machine-to-Machine (M2M) economy** in which agents own wallets and budgets, execute millions of micro-transactions, carry **identity and reputation** (Agent Passport), and use **delegated** and **streaming** payments. AIFP is to payments what TLS became to security and OAuth to authorization: it makes **payment a native part of HTTP semantics** for the machine web, finally activating the `402 Payment Required` status code that HTTP/1.1 reserved for exactly this purpose.

2. Design Goals

An AIFP-1 implementation is designed against eight principles. These are normative design constraints; conformance requirements derived from them appear in Section 23.

1. **Zero friction for machines.** Payment **MUST** complete without human involvement once an agent is funded.
2. **Stateless verification.** A merchant **MUST** be able to verify a receipt locally by signature, without contacting the AiFinPay backend.
3. **HTTP-native.** The protocol **MUST** rely on standard HTTP semantics — 402 , standard headers, idempotency.
4. **x402 compatibility.** Where appropriate, AIFP **SHOULD** interoperate with the x402 ecosystem.
5. **Multi-chain and fiat-agnostic.** Settlement **MAY** occur in stablecoins across 12 supported networks or in fiat.
6. **Secure by default.** Replay, double-spend, and MITM protections **MUST** be built in, not optional add-ons.
7. **Enterprise scale.** The architecture **MUST** sustain millions of requests per second and billions of receipts per day.
8. **Open protocol.** The specification **SHOULD** remain open; implementations **MAY** compete.

3. Terminology

The full glossary is in **Appendix A**. The terms below are sufficient to read this document standalone.

- **Agent.** An autonomous software client (LLM agent, crawler, pipeline) that consumes resources and pays for them.
- **Merchant.** A provider of a paid resource (API or content) that integrates AIFP to monetize machine traffic.
- **Wallet.** A funding source bound to an agent: an on-chain wallet (custodial, non-custodial, or MPC) or a fiat balance.
- **Quote.** A server-issued, time-bounded price for accessing a specific resource at a specific pricing_tier.
- **Payment Challenge.** The machine-readable 402 payload telling an agent how and how much to pay.
- **Receipt Token.** A signed, time-bounded cryptographic proof of payment that grants access on retry.
- **Nonce.** A single-use value embedded in a challenge and receipt to prevent replay.
- **Free Quota.** The number of requests a merchant serves free before charging (default 100).
- **Pricing Tier Tier.** A merchant-assigned cost class for a resource: *standard / standard / complex / premium*.
- **Settlement.** The act of moving value to the merchant on-chain or via fiat rails.
- **Stateless Verification.** Local receipt validation by signature, exp, nonce, and amount, with no backend call.
- **Control Plane / Data Plane.** AiFinPay backend (quoting, payment, receipts) vs. merchant middleware (interception, verification).

4. Protocol Overview

4.1. Architectural model

AIFP separates a strongly consistent **Control Plane** (the AiFinPay backend: quoting, payment execution, receipt issuance, ledger, webhooks) from a replicated, low-latency **Data Plane** (the merchant middleware: request interception, quota checks, stateless receipt verification). This separation is the foundation of AIFP's scalability: **receipts are verified on the merchant side by signature**, so the backend is never in the per-request hot path.



4.2. Why stateless verification matters

Ed25519 signature verification runs at roughly **50,000 verifications per second per CPU core**. By verifying receipts locally, a merchant can authorize millions of paid requests per second with no dependence on AiFinPay availability. The backend's job ends the moment a receipt is signed; everything after is merchant-local. This is the single most important architectural decision in AIFP-1.

4.3. Degraded mode

Because verification is local, a merchant **MUST** continue to honor valid, unexpired receipts **even when the AiFinPay Control Plane is unreachable**. The Data Plane is eventually consistent and **MUST** keep working in degraded mode. New payments may be unavailable during a Control Plane outage, but already-issued receipts remain valid until expiry.

5. HTTP 402 Semantics

5.1. Activating a reserved status code

HTTP/1.1 ([RFC 9110]) reserves **402 Payment Required** "for future use." AIFP-1 defines that use. A merchant **MUST** return **402** when, and only when, access to a resource requires payment that has not yet been provided — specifically, when an identified agent's free quota is exhausted and no valid receipt accompanies the request.

5.2. Normative rules for 402

- A 402 response MUST carry a machine-readable Payment Challenge (Section 6), in a header, a body, or both.
- A 402 response MUST be **idempotent and side-effect free**: issuing a challenge MUST NOT consume quota or charge the agent.
- A 402 response MUST include a `nonce` and an `expires_at` so the challenge cannot be replayed indefinitely.
- A merchant MUST NOT return 402 for requests that carry a **valid, unexpired, correctly scoped** receipt; such requests MUST be served (or fail for an unrelated reason).
- A bare 402 with no challenge is **non-conformant**. When an agent does not advertise AIFP support, the merchant MUST return the onboarding form of 402 (Section 15).

5.3. Distinction from neighboring codes

Code	Meaning in AIFP	Not to be confused with
402	Payment required to proceed	The protocol core
403	Agent blocklisted / merchant KYC fail	A <i>*policy*</i> denial, not a price
409	Receipt replay / idempotency conflict	A <i>*uniqueness*</i> failure
425	Payment not yet settled on-chain	Retry after <code>Retry-After</code> , same receipt

6. Payment Challenge

6.1. Purpose

The Payment Challenge is the machine-readable instruction set an agent needs to pay: **where** to get a quote, **what** resource, **which** pricing_tier tier, **how much** (estimate), **which** assets and chains are accepted, a single-use **nonce**, and an **expiry**.

6.2. Variant A — header form (x402-compatible)

```
HTTP/1.1 402 Payment Required
WWW-Authenticate: AIFP realms="api.example.com", quote="https://api.aifinpay.io/v1/quote",
  endpoint="/api/data", pricing_tier="standard"
AIFP-Payment-Required: true
Accept-Payment: aifp/1.0
Content-Type: application/json
```

6.3. Variant B — body form (canonical)

```
{
  "error": {
    "type": "payment_required",
    "code": "AIFP-402",
    "message": "Free quota exhausted. Payment required to continue."
  },
  "payment_challenge": {
    "version": "1.0",
    "scheme": "aifp",
    "quote_endpoint": "https://api.aifinpay.io/v1/quote",
    "merchant_id": "mrch_9f3a1c2b",
    "resource": "/api/data",
    "pricing_tier": "standard",
    "estimated_amount": "0.04",
    "currency": "USD",
    "accepted_assets": [ "USDC", "USDT", "PYUSD" ],
    "accepted_chains": [ "polygon", "base", "solana", "unichain" ],
    "nonce": "b7e2...c91a",
    "expires_at": "2026-06-28T12:34:56Z"
  }
}
```

6.4. Field requirements

Field	Req.	Notes			
scheme	MUST	Always "aifp" for this protocol			
quote_endpoint	MUST	Where the agent requests a binding quote			
merchant_id	MUST	Stable merchant identifier mrch_*			
resource	MUST	The protected path/resource			
pricing_tier	MUST	standard \	standard \	complex \	premium
estimated_amount	SHOULD	Decimal string, USD-denominated estimate			
currency	SHOULD	ISO-4217, default "USD"			
accepted_assets	SHOULD	Stablecoins accepted, e.g. USDC , USDT , PYUSD			
accepted_chains	SHOULD	Networks accepted (Appendix B)			
nonce	MUST	Single-use, ≥128 bits entropy			
expires_at	MUST	RFC 3339 timestamp			

6.5. Pricing tiers (canonical)

Pricing Tier	Price (USD)	Typical use
standard	USD 0.00001	Typical API call, single record

Pricing Tier	Price (USD)	Typical use
complex	USD 0.00006	Aggregations, search, multi-record
premium	USD 0.00010	Inference, heavy compute, premium data

Free quota default: **100 requests** per agent before charging begins. Both pricing and quota are merchant-configurable; the values above are the protocol defaults and **MUST** be used when unset.

7. Receipt Token

7.1. Definition

A **Receipt Token** is a signed, time-bounded, audience-bound cryptographic proof that a specific agent paid a specific merchant for a specific resource. It is the credential that turns a `402` into a `200` on retry. Receipts are **bearer-restricted**: bound to an audience and a nonce so they cannot be replayed or reused across resources.

7.2. Encoding

A Receipt Token is a **JWS-signed JWT** (compact serialization) using **EdDSA / Ed25519**. A compact **CBOR-COSE** variant is defined for constrained environments (Section 16.4). Both carry identical claims.

7.3. Claims



Claim	Meaning
sub	The paying agent (<code>agt_*</code>)
aud	The merchant the receipt is valid for (<code>mrch_*</code>) — verification MUST reject a mismatch

Claim	Meaning
resource	Resource paid for — MUST match the retried request
amount / currency / asset / chain	Settlement details
tx_ref	On-chain tx hash or fiat settlement reference
receipt_id	Unique receipt identifier
nonce	Single-use anti-replay value
iat / exp	Issued-at / expiry (default TTL 600 s)
kid	Key ID for signature verification & rotation

7.4. Stateless verification algorithm

A merchant **MUST** verify a receipt locally, with no network call, as follows:

```

1. Parse the JWT and read 'kid'.
2. Resolve the issuer public key for 'kid' (from a cached JWKS).
3. Verify the EdDSA signature. On failure → 422.
4. Check 'iss' is the trusted AiFinPay issuer. On failure → 422.
5. Check 'aud == this merchant id'. On failure → 403.
6. Check 'resource == requested resource'. On failure → 422.
7. Check 'now < exp' (with ≤30s clock skew). On failure → 402 (expired).
8. Check 'amount >= required price' for the resource/pricing tier. On failure → 422.
9. Check 'nonce' has NOT been seen before (local nonce store). On replay → 409.
10. Mark 'nonce' used (TTL = exp). Serve 200. Done.

```

Steps 1–8 are pure and stateless. Step 9 requires a small, fast **nonce store** (Redis or in-memory with TTL) sized to the receipt TTL window — typically minutes, not forever. See Section 18.5.

7.5. Receipt lifecycle

```

stateDiagram-v2
    [*] --> Issued: payment settled
    Issued --> Redeemed: presented & verified
    Issued --> Expired: exp passed unredeemed
    Redeemed --> [*]
    Expired --> [*]
    Issued --> Revoked: fraud / chargeback
    Revoked --> [*]

```

A receipt is **single-use** by default (nonce consumed on redemption). Merchants **MAY** configure a receipt as multi-use within its TTL for a single resource (e.g., paginated reads) by issuing a `quota` claim; absent that claim, one receipt grants one access.

8. HTTP Specification — Headers, Versioning, Idempotency

8.1. Base URLs

```
production: https://api.aifinpay.io/
sandbox: https://sandbox.api.aifinpay.io/v
```

8.2. Request headers

Header	Required	Description
Idempotency-Key: <uuid>	mutations	POST idempotency
AIFP-Version: 1.0	SHOULD	Protocol version
Content-Type: application/json	yes	Body format
AIFP-Agent-ID: <agent_id>	agent calls	Agent identifier
Accept-Payment: aifp/1.0	agent	Advertises AIFP support
Payment-Receipt: <token>	on retry	Receipt Token

8.3. Response headers

Header	Description
AIFP-Version	Negotiated protocol version
Retry-After	Seconds before retry (429 / 425 / 503)
RateLimit-Limit / RateLimit-Remaining / RateLimit-Reset	Rate-limit signaling

8.4. Versioning

Versioning is by path prefix (/v1). Backward-compatible additions do not change the version. Breaking changes introduce /v2 . Clients SHOULD send AIFP-Version ; servers MUST echo the negotiated version. A server MUST NOT silently break a documented /v1 contract.

8.5. Idempotency

Every state-changing POST (notably /pay) MUST accept an Idempotency-Key . The server MUST store the first response keyed by (api_key, Idempotency-Key) for **24 hours** and return the **identical** response for any repeat with the same key and body. A repeat with the same key but a **different body** MUST return 409

Conflict . This makes payment retries safe: a client that times out can safely re- POST /pay and will never double-charge.

9. Complete Protocol Flow

9.1. Happy path

Step	Actor	Action	HTTP
2	Merchant SDK	Identify agent, check quota	—
3a	Merchant SDK	Quota available → serve	200 OK
3b	Merchant SDK	Quota exhausted → challenge	402 + challenge
4	Agent	Parse challenge	—
5	Agent	Request quote	POST /quote
6	Agent	Pay	POST /pay + Idempotency-Key
7	Payment Engine	Execute & settle	—
8	Receipt Service	Issue Receipt Token	—
9	Agent	Retry with receipt	GET /api/data + Payment-Receipt
10	Merchant SDK	Verify locally (Ed25519)	—
11	Merchant	Grant access	200 OK + data

9.2. Sequence diagram



```
B-->>P: tx hash / settlement id
P->>R: Request receipt
R-->>G: signed Receipt Token
G-->>A: 200 {receipt, receipt_id, expires_at}
A->>M: GET /api/data + Payment-Receipt: <token>
M->>M: Verify(sig, aud, resource, exp, amount, nonce)
alt Receipt valid
M->>M: Mark nonce used
M-->>A: 200 OK (data)
else Receipt invalid
M-->>A: 402 / 409 (replay) / 42
end
end
```

9.3. Timeouts

Timeout	Default	Behavior
Payment settlement	30 s fiat / chain-dependent on-chain	202 + webhook if exceeded
Receipt TTL	600 s	Receipt invalid after expiry
Idempotency window	24 h	Key retained 24 h
Connection timeout	10 s	Connection dropped

9.4. Webhooks

AiFinPay emits signed webhooks on payment lifecycle events: `payment.succeeded` , `payment.pending` , `payment.failed` , `receipt.issued` , `receipt.redeemed` , `settlement.completed` , `payout.completed` , `dispute.opened` . Each is signed `AIFP-Signature: t=<ts>,v1=<hmac>` (HMAC-SHA256). The merchant MUST verify signature and timestamp (5-minute replay window). Delivery is at-least-once with exponential retry up to 72 hours, then dead-letter.

10. Authentication & Authorization

10.1. Two credential planes

AIFP distinguishes **identity** (who you are) from **payment** (proof you paid):

- **API keys** authenticate control-plane calls (`/quote` , `/pay` , dashboard, webhooks). Format: `sk_live_*` (secret) and `pk_live_*` (publishable). Keys are scoped (merchant vs. agent) and rotatable.
- **Receipt Tokens** authorize **resource access** on the data plane. A valid receipt is sufficient to access a resource; it does not require an API key on the retried request.

10.2. Agent identity

An agent is identified by `AIFP-Agent-ID` and, optionally, an **Agent Passport** (a signed identity credential, `agt_* / pp_*`, Ed25519). The Passport binds an agent to its wallets, budget policies, and reputation. Full Passport semantics are defined in the [AI Agent SDK Specification](#) and the [Security Specification](#). For AIFP-1, a Passport is OPTIONAL: an agent MAY pay anonymously with only a funded wallet.

10.3. Authorization rules

- A merchant MUST grant access to any request carrying a valid receipt scoped to that merchant and resource.
- A merchant MAY blocklist an `agt_*` (returning `403`) independently of payment.
- A merchant MUST NOT require human registration of an agent as a condition of payment.

11. Merchant Flow

```
stateDiagram-v2
    [*] --> Receive: HTTP request arrives
    Receive --> Identify: extract Agent-ID / fingerprint
    Identify --> QuotaCheck
    QuotaCheck --> Serve: quota remaining
    QuotaCheck --> ReceiptCheck: quota exhausted
    ReceiptCheck --> Verify: Payment-Receipt present
    ReceiptCheck --> Challenge: no receipt
    Verify --> Serve: valid
    Verify --> Replay: nonce reused (409)
    Verify --> Reject: invalid (422)
    Challenge --> [*]: 402 + challenge
    Serve --> [*]: 200 + data
```

The merchant integrates AIFP as **middleware** (Data Plane). On each request it (1) identifies the agent, (2) checks the free quota, (3) if quota remains, serves and decrements, (4) else checks for a receipt and verifies it locally, (5) on success serves the resource, (6) otherwise returns a `402` challenge. Concrete, production-ready integrations for 15 frameworks are in the [Merchant Integration Guide](#).

12. Agent Flow

```
stateDiagram-v2
    [*] --> Request
    Request --> Got200: 200 OK
    Request --> Got402: 402 Payment Required
    Got402 --> Quote: POST /quote
    Quote --> Pay: POST /pay (Idempotency-key)
    Pay --> Settling: 202 settling
    Pay --> Receipt: 200 receipt
    Settling --> Receipt: webhook / post
    Receipt --> Retry: replay original + receipt
```

Retry --> Got200

Got200 --> [*]

An AIFP agent: sends a request; on `402`, parses the challenge, gets a quote, pays with an idempotency key, obtains a receipt, and **replays the original request** with `Payment-Receipt`. The agent **MUST** implement budget controls, retry with backoff, and receipt caching. The full SDK contract — quote/pay/verify APIs, budget policies, multi-wallet routing, state machine — is in the [AI Agent SDK Specification](#).

13. Wallet Flow

13.1. Wallet types

Type	Custody	Use
Non-custodial	Agent-held key	Agent controls keys; signs locally
MPC	Threshold-shared	No single key; enterprise-grade
Fiat balance	AiFinPay ledger	Prepaid USD balance, fiat/stablecoin settlement

13.2. Funding and budgets

A wallet is funded on-chain (stablecoin deposit) or via fiat top-up. Each wallet carries a **budget policy**: per-request cap, daily/monthly caps, per-merchant allowances, and an optional auto-refill. The protocol error `AIFP-403-BUDGET-EXCEEDED` is returned when a payment would breach a budget policy. Wallet binding to an Agent Passport, multi-wallet routing, and MPC operation are specified in Documents 3 and 4.

14. x402 Compatibility & Migration

14.1. Compatibility

AIFP-1 is designed to interoperate with the **x402** ecosystem, which also builds on `402 Payment Required`. AIFP advertises support via `Accept-Payment: aifp/1.0` and the `WWW-Authenticate: AIFP ...` header (Section 6.2). A merchant **MAY** accept both x402 and AIFP challenges on the same endpoint; an agent **MAY** negotiate the scheme via the Protocol Negotiation Layer (Section 22.3).

14.2. Migration

An agent that already supports x402 can migrate to AIFP in **one click** and receives a **1,000 free paid-request** bonus. The migration endpoint exchanges an existing x402 identity for an AIFP agent identity and wallet binding, preserving the agent's request flow. After migration, the agent uses AIFP receipts and gains access to budgets, multi-chain routing, and (optionally) an Agent Passport.

```
POST /v1/migrate/x402
Authorization: Bearer <agent_api_key>
Content-Type: application/json
```

```
{ "x402_identity": "...", "preferred_chain": "polygon", "preferred_asset": "USDC"
```

```
{ "agent_id": "agt_4f9a2c7e", "wallet_id": "wlt ...", "free_requests_granted": 1000, "migrated": true }
```

15. Onboarding Flow

A **402** MUST NOT be a dead end. When an agent does **not** advertise AIFP support (`Accept-Payment` absent), the merchant MUST return an **onboarding** **402** that explains what AiFinPay is, why payment is required, and exactly how to connect — comparable in polish to Stripe Connect.

```
{
  "error": { "type": "payment_required", "code": "AIFP-402-ONBOARDING" },
  "message": "This resource is monetized via AiFinPay. Your agent does not yet support AIFP. ",
  "payment_challenge": { "version": "1.0", "scheme": "aifp", "quote_endpoint":
    "https://api.aifinpay.io/v1/quote",
    "merchant_id": "mrch_9f3a1c2b", "resource": "/api/data", "pricing_tier": "standard",
    "estimated_amount": "0.04", "currency": "USD", "nonce": "b7e2...c91a", "expires_at": "2026-06-
    18T12:34:56Z" },
  "agent_supported": false,
  "onboarding": {
    "what_is_aifinpay": "AiFinPay is an application-layer payment protocol on top of HTTP that lets AI
    agents automatically pay for access to content and APIs without human involvement.",
    "why_payment_required": "The free quota (100 requests) is exhausted. Further access to this resource is
    paid.",
    "actions": {
      "connect_wallet": "https://app.aifinpay.io/wallet/connect",
      "create_wallet": "https://app.aifinpay.io/wallet/create",
      "fund_wallet": "https://app.aifinpay.io/wallet/fund",
      "install_sdk": "npm install @aifinpay/agent",
      "developer_docs": "https://docs.aifinpay.io/agents"
    },
    "x402_migration": { "available": true, "detail": "One-click migration with a 1000 free paid-request
    bonus. See Section 14." }
  }
}
```

When the agent **does** advertise support (`Accept-Payment: aifp/1.0`), the merchant returns the automatic challenge (Section 6.3) with `"agent_supported": true` and no onboarding block.

16. Message Formats

16.1. POST /v1/quote

Request

```
{ "merchant_id": "mrch_9f3alc2b", "resource": "/api/data", "pricing_tier": "standard", "currency": "USD" }
```

Response 200

```
{
  "quote_id": "qt_8d21f0",
  "merchant_id": "mrch_9f3alc2b",
  "resource": "/api/data",
  "amount": "0.04",
  "currency": "USD",
  "accepted_assets": [ "USDC", "USDT", "PYUSD" ],
  "accepted_chains": [ "polygon", "base", "solana" ],
  "pay_to": { "polygon": "0xMerchant...", "solana": "Merc...111" },
  "nonce": "b/e2...c91a",
  "expires_at": "2026-06-28T12:39:56Z"
}
```

16.2. POST /v1/pay

Request (header Idempotency-Key: <uuid>)

```
{ "quote_id": "qt_8d21f0", "wallet_id": "wit_3a1b", "asset": "USDC", "chain": "polygon" }
```

Response 200

```
{
  "receipt_id": "rcpt_7b3e9f21",
  "receipt": "<JWT EdDSA receipt token>",
  "status": "settled",
  "tx_ref": "0xabcd123...",
  "amount": "0.04",
  "fee": "0.00012",
  "expires_at": "2026-06-28T12:44:56Z"
}
```

Response 202 (async settlement)

```
{ "receipt_id": "rcpt_7b3e9f21", "status": "settling", "poll": "/v1/receipt/rcpt_7b3e9f21" }
```

16.3. GET /v1/receipt/{receipt_id}

Returns the receipt and its status. Used when settlement was async (202).

16.4. CBOR-COSE receipt (constrained clients)

For embedded or bandwidth-constrained agents, a receipt MAY be encoded as a **CBOR Web Token (CWT)** signed with **COSE / EdDSA**. Claim semantics are identical to Section 7.3; the encoding is compact binary. A merchant that advertises `cwt` capability (Section 22) MUST accept either form.

16.5. Fee model

AIFP charges a volume-tiered fee on top of the resource price: **Tier 1 1% / Tier 2 1% / Tier 3 1%** (capped at ~1%). The fee is computed and shown at quote time and itemized on the receipt (`fee` field).

17. Error Codes & Retry Strategy

17.1. Status code matrix

Code	AIFP semantics	When
202	Accepted (async)	Payment processing, async settlement
400	Format error	Invalid JSON / missing field
401	Unauthorized	Missing/invalid API key
402	Payment required	Quota exhausted; protocol core
403	Forbidden	Agent blocklisted / KYC fail / budget exceeded
404	Not found	<code>quote_id</code> / <code>receipt_id</code> unknown
409	Conflict	Receipt replay / idempotency reuse w/ different body
410	Gone	Quote expired
422	Unprocessable	Amount mismatch / bad signature
425	Too early	Payment not yet settled on-chain
429	Rate limited	Request limit exceeded
500/502/503/504	Server/gateway	Internal / upstream / maintenance / timeout

17.2. Structured error body

```
{
  "error": {
    "type": "invalid_receipt",
    "code": "AIFP-422-SIGNATURE",
    "message": "Receipt signature verification failed.",
    "request_id": "req_8f12",
    "doc": "https://docs.aifinpay.io/errors/AIFP-422-SIGNATURE"
  }
}
```

The complete error registry is in **Appendix C**.

17.3. Retry strategy

Agents MUST use exponential backoff with jitter:

```
delay = min(base * 2^attempt + jitter, max_delay)
base = 200ms, max_delay = 30s, max_attempts = 5
```

- On `402`, the agent MUST NOT retry without paying.
- On `425`, the agent MUST honor `Retry-After` and retry with the **same** `receipt_id`.
- On `429`, the agent MUST honor `Retry-After`.
- On `5xx`, the agent MAY retry idempotently with the **same** `Idempotency-Key`.
- On `409` (replay), the agent MUST obtain a **fresh** quote/receipt; it MUST NOT reuse the consumed nonce.

18. Security Model & Threat Model

This section is normative-summary. The exhaustive treatment — cryptographic primitives, key rotation, MPC, fraud detection, OWASP/SOC2/ISO mappings — is in the [Security & Cryptography Specification](#).

18.1. Trust model

- The **AiFinPay issuer key** is the root of receipt trust. Merchants trust receipts signed by a current `kid` in the published JWKS.
- Merchants and agents authenticate to the Control Plane with **API keys**.
- No party trusts the network: every security-critical artifact (receipt, webhook) is **signed**.

18.2. Threat model (STRIDE summary)

Threat	Vector	Mitigation
Tampering	Modified challenge/receipt	Signed receipts; HMAC-signed webhooks
Repudiation	"I never paid"	<code>tx_ref</code> on-chain proof + signed receipt audit log
Information disclosure	Sniffing	TLS 1.3 mandatory; no secrets in receipts
Denial of service	Request flood	Rate limiting, anycast, local verification
Elevation of privilege	Replay / reuse	Single-use nonce; idempotency keys

18.3. Replay protection

Every challenge and receipt carries a **single-use nonce** (≥ 128 bits). The merchant maintains a nonce store with TTL equal to the receipt TTL. A re-presented nonce MUST be rejected with `409 Conflict`. Because TTL is short (minutes), the nonce store stays small even at billions of receipts/day.

18.4. Cryptography

- **Signatures:** EdDSA / **Ed25519** (receipts), HMAC-SHA256 (webhooks).
- **Transport:** TLS 1.3, MUST.
- **Hashing:** SHA-256 / SHA-512.
- **Encoding:** JWT (JOSE) and CWT (COSE).
- **Key rotation:** receipts carry `kid`; issuer publishes a rotating JWKS; merchants cache and refresh.

18.5. Idempotency & double-spend

Idempotency-Key on `/pay` prevents double-charge on retry. On-chain `tx_ref` plus the consumed nonce prevents a receipt being spent twice. The combination guarantees **at-most-once** value transfer per logical payment.

19. Settlement, Multi-chain & Fiat

19.1. Settlement models

- **On-chain stablecoin.** The agent's wallet pays the merchant address in `USDC / USDT / PYUSD`; `tx_ref` is the tx hash. A non-custodial on-chain **payment splitter** can route protocol fees and multi-party splits atomically.
- **Fiat / stablecoin hybrid.** Via fiat rails (e.g., BVNK), settlement may originate or terminate in fiat with stablecoin in between. The receipt's `tx_ref` carries the settlement reference.

19.2. Supported networks (12)

Networks are organized in three capability tiers (full list in **Appendix B**):

- **Full Core (8):** Solana, Polygon, Avalanche, BNB Chain, Optimism, Arbitrum, Base, Unichain — Core + Passport + mSECCO escrow + Pyth oracle.
- **Splitter-only EVM (2):** BOT Chain, XRPL EVM.
- **Splitter MVP non-EVM (2):** NEAR, Aptos.

19.3. Settlement state machine

```
stateDiagram-v2
    [*] --> Initiated: POST /pay
    Initiated --> Pending: broadcast / fiat submit
    Pending --> Settled: confirmed
    Pending --> Failed: reverted / declined
    Settled --> Receipted: receipt issued
    Failed --> [*]
    Receipted --> [*]
```

A merchant **MUST NOT** require settlement **confirmation** before accepting a receipt: the receipt itself is the access credential, and AiFinPay assumes settlement risk for issued receipts. Where confirmation is required (high-value resources), the merchant **MAY** demand 425 -then- 200 semantics keyed on `tx_ref`.

20. State Machines

The protocol's three core state machines — **Merchant** (Section 11), **Agent** (Section 12), and **Settlement** (Section 19.3) — together with the **Receipt lifecycle** (Section 7.5) fully describe AIFP-1 behavior. An implementation **MUST NOT** introduce states that allow value transfer without a corresponding receipt, nor resource access without receipt verification.

21. OpenAPI & API Examples

21.1. OpenAPI 3.1 (excerpt)

```
openapi: 3.1.0
info:
  title: AiFinPay Payment Protocol API
  version: 1.0.0
servers:
  - url: https://api.aifinpay.io/v1
paths:
  /quote:
    post:
      summary: Create a binding quote
      requestBody:
        required: true
        content:
          application/json:
            schema: { $ref: '#/components/schemas/QuoteRequest' }
      responses:
        '200': { description: Quote created, content: { application/json: { schema: { $ref: '#/components/schemas/Quote' } } } }
        '410': { description: Resource pricing unavailable }
  /pay:
    post:
      summary: Execute a payment against a quote
      parameters:
        - in: header
          name: Idempotency-Key
          required: true
          schema: { type: string, format: uuid }
      requestBody:
        required: true
        content:
          application/json:
            schema: { $ref: '#/components/schemas/PayRequest' }
      responses:
        '200': { description: Settled, content: { application/json: { schema: { $ref: '#/components/schemas/Payment' } } } }
```

```

    '#/components/schemas/Receipt' } } } }
    '202': { description: Settling
    '409': { description: Idempotency conflict
components:
  schemas:
    QuoteRequest:
      type: object
      required: [merchant_id, resource, pricing_tier]
      properties:
        merchant_id: { type: string }
        resource: { type: string }
        pricing_tier: { type: string, enum: [standard, standard, complex, premium] }
        currency: { type: string, default: USD }
    Quote:
      type: object
      properties:
        quote_id: { type: string }
        amount: { type: string }
        currency: { type: string }
        accepted_assets: { type: array, items: { type: string } }
        accepted_chains: { type: array, items: { type: string } }
        nonce: { type: string }
        expires_at: { type: string, format: date-time }
    PayRequest:
      type: object
      required: [quote_id, wallet_id]
      properties:
        quote_id: { type: string }
        wallet_id: { type: string }
        asset: { type: string }
        chain: { type: string }
    Receipt:
      type: object
      properties:
        receipt_id: { type: string }
        receipt: { type: string, description: EdDSA JWT }
        status: { type: string, enum: [settled, settling, failed] }
        tx_ref: { type: string }
        amount: { type: string }
        fee: { type: string }
        expires_at: { type: string, format: date-time }

```

21.2. cURL — end-to-end

```

# 1. Hit a protected resource (quota exhausted) → 402
curl -i https://merchant.example.com/api/data -H "Accept-Payment: aifp/1.0"

# 2. Quote
curl -s https://api.aifinpay.io/v1/quote \
  -H "Authorization: Bearer $AGENT_KEY" -H "Content-Type: application/json" \
  -d '{"merchant_id":"mrch_9f3a1c2b","resource":"/api/data","pricing_tier":"standard"}'

# 3. Pay (idempotent)
curl -s https://api.aifinpay.io/v1/pay \
  -H "Authorization: Bearer $AGENT_KEY" -H "Idempotency-Key: ${uuidgen}" \
  -H "Content-Type: application/json" \
  -d '{"quote_id":"qt_8d21f0","wallet_id":"wlt_3a1b","asset":"USDC","chain":"polygon"}'

```

```
4. Retry the original request with the receipt
url -i https://merchant.example.com/api/data
-H "Payment-Receipt: $RECEIPT_JWT"
```

22. Open Governance & Versioning

22.1. Governance

AIFP evolves through an **open RFC process** (AIFP-N). Anyone may submit a proposal; proposals move through *Draft → Review → Last Call → Accepted/Rejected → Final*. Reference implementations, compliance tests, and a certification suite accompany each ratified change.

22.2. Versioning policy

- **Patch / minor:** backward-compatible; same /v1 , advertised via AIFP-Version .
- **Major:** breaking; new path prefix (/v2); migration guide REQUIRED.
- Servers MUST support at least the two most recent minor versions.

22.3. Protocol Negotiation Layer

Clients and servers MAY negotiate capabilities (receipt encoding jwt / cwt , accepted assets/chains, optional features) via capability discovery (GET /.well-known/aifp) and request/response headers. A server MUST fall back gracefully to baseline AIFP-1 when a client requests no extensions.

23. Normative Requirements Summary

An implementation claiming **AIFP-1 conformance** MUST satisfy all of the following:

R1 — A merchant MUST return 402 with a machine-readable Payment Challenge (Section 6) when, and only when, an identified agent's quota is exhausted and no valid receipt is present.

R2 — A 402 MUST be side-effect free and MUST carry a single-use nonce and expires_at .

R3 — A merchant MUST verify receipts **locally** by Ed25519 signature, checking iss , aud , resource , exp , amount , and nonce (Section 7.4).

R4 — A merchant MUST reject a reused nonce with 409 and MUST keep working (degraded) when the Control Plane is unavailable.

R5 — Receipts MUST be EdDSA-signed JWTs (or COSE-CWT equivalents) carrying the claims of Section 7.3, with default TTL 600 s.

R6 — /pay MUST honor Idempotency-Key for 24 h and MUST NOT double-charge.

- R7** — Default pricing (Standard from USD 0.00001 / Complex from USD 0.00006 / Premium from USD 0.00010) and free quota (100) MUST be used when unconfigured.
- R8** — Agents MUST implement exponential backoff with jitter and MUST NOT retry a 402 without paying.
- R9** — TLS 1.3 MUST be used for all control-plane traffic; webhooks MUST be HMAC-signed and timestamp-verified.
- R10** — A bare 402 (no challenge) is non-conformant; absent agent support, the onboarding 402 (Section 15) MUST be returned.

24. Future Extensions

AIFP-1 is forward-compatible with the following extensions, specified in companion documents and governed by Section 22:

- **Agent Passport** — portable signed agent identity (agt_* / pp_*), wallet binding, delegated spending.
- **Merchant Discovery Registry** — a registry and .well-known self-description so agents can discover priced resources.
- **Dynamic Pricing Engine** — rule-based pricing clamped to [min, max] , with reputation discounts.
- **Streaming Payments** — payment channels for continuous/metered access.
- **Agent Reputation Network** — reputation ∈ [0,1000] and risk ∈ [0,100] influencing pricing and trust.

None of these are required for AIFP-1 conformance; all are optional and capability-negotiated.

Appendix A. Glossary

Term	Definition
Agent	Autonomous software client that consumes and pays for resources.
Agent Passport	Optional signed agent identity credential (agt_* / pp_*).
Pricing Tier Tier	Cost class of a resource: standard/standard/complex/premium.
Control Plane	AiFinPay backend: quoting, payment, receipts, ledger, webhooks.
Data Plane	Merchant middleware: interception, quota, local verification.
Free Quota	Free requests served before charging (default 100).
Idempotency-Key	Client-supplied key making POST /pay safe to retry.
JWKS	JSON Web Key Set — issuer public keys for receipt verification.
kid	Key ID identifying the signing key for rotation.
Merchant	Provider of a paid resource integrating AIFP.
Nonce	Single-use anti-replay value in challenge and receipt.

Term	Definition
Payment Challenge	Machine-readable 402 payload instructing how to pay.
Quote	Time-bounded server-issued price for a resource.
Receipt Token	Signed time-bounded proof of payment granting access.
Settlement	Movement of value to the merchant on-chain or via fiat.
Stateless Verification	Local receipt validation with no backend call.
x402	Adjacent 402 -based payment ecosystem AIFP interoperates with.

Appendix B. Supported Networks

Tier	Networks	Capabilities
Splitter-only EVM (2)	BOT Chain, XRPL EVM	Payment splitter
Splitter MVP non-EVM (2)	NEAR, Aptos	Payment splitter (MVP)

Accepted assets: USDC , USDT , PYUSD (network-dependent). Fiat/stablecoin hybrid settlement via BVNK.

Appendix C. Complete Error Registry

Code	HTTP	Meaning
AIFP-401	401	Missing/invalid API key
AIFP-402	402	Payment required (challenge)
AIFP-402-ONBOARDING	402	Payment required + onboarding (agent lacks AIFP)
AIFP-403	403	Forbidden (blocklist / KYC)
AIFP-403-BUDGET-EXCEEDED	403	Payment would breach budget policy
AIFP-404	404	Quote/receipt not found
AIFP-409	409	Receipt replay / idempotency conflict
AIFP-410	410	Quote expired
AIFP-422-SIGNATURE	422	Receipt signature invalid
AIFP-422-AMOUNT	422	Amount below required price
AIFP-425	425	Settlement not yet confirmed
AIFP-429	429	Rate limited
AIFP-5xx	5xx	Server/gateway/timeout

Appendix D. Changelog

Version	Date	Change
---------	------	--------

Appendix E. References

- [RFC 2119] Key words for use in RFCs to Indicate Requirement Levels.
- [RFC 8174] Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words.
- [RFC 9110] HTTP Semantics.
- [RFC 7519] JSON Web Token (JWT).
- [RFC 8037] CFRG EdDSA in JOSE (Ed25519).
- [RFC 8949] Concise Binary Object Representation (CBOR).
- [RFC 8392] CBOR Web Token (CWT).
- [RFC 9052] CBOR Object Signing and Encryption (COSE).
- x402 — HTTP 402 payment ecosystem.
- Companion documents: Merchant Integration Guide; AI Agent SDK Specification; Security & Cryptography Specification.

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